

To find the number of "good" polynomials, we analyze the given conditions. Let the polynomial be

$$P(x) = x^7 + a_6x^6 + a_5x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0$$

Since P is monic, its leading coefficient is $a_7 = 1$. The conditions state that the absolute values of all coefficients of P are odd, strictly less than 2025^2 , and are roots of P . Let $S = \{|a_0|, |a_1|, |a_2|, |a_3|, |a_4|, |a_5|, |a_6|, |a_7|\}$. Since $a_7 = 1$, $1 \in S$, so 1 must be a root of P . Because the roots of a monic polynomial with integer coefficients are algebraic integers, the product of any subset of its roots is also an algebraic integer. The product of all roots of P is $-a_0$, so the absolute value of the product of all roots is $|a_0|$.

Let $A = |a_0|$. Since $A \in S$, A is an integer root of P . If S contains another integer $B > 1$ distinct from A , B would also be a root of P . Then the product of the remaining roots would be $\pm A/(A \cdot B) = \pm 1/B$. However, since $B > 1$, $\pm 1/B$ is a rational number that is not an integer, so it cannot be an algebraic integer. This yields a contradiction. Similarly, A cannot be a multiple root. Thus, S can only consist of 1 and A . This means every coefficient's absolute value is either 1 or A .

Case 1: $A = 1$ If $A = 1$, then $|a_i| = 1$ for all $i = 0, \dots, 7$. The absolute values are all 1, which is odd and strictly less than 2025^2 . The only requirement is that 1 is a root of P , meaning $P(1) = 0$. Evaluating at $x = 1$ gives:

$$1 + a_6 + a_5 + a_4 + a_3 + a_2 + a_1 + a_0 = 0$$

Since each coefficient is ± 1 , exactly four must be $+1$ and four must be -1 . Since $a_7 = 1$ is positive, exactly three of the remaining seven coefficients must be $+1$. The number of such polynomials is:

$$\binom{7}{3} = 35$$

Case 2: $A \geq 3$ If $A \geq 3$, since A is a root, $P(x)$ factors as $P(x) = (x - A)Q(x)$, where $Q(x)$ is a monic polynomial of degree 6 with integer coefficients. The roots of $Q(x)$ must be algebraic units (meaning the constant term of $Q(x)$ is ± 1). Let $Q(x) = x^6 + c_5x^5 + c_4x^4 + c_3x^3 + c_2x^2 + c_1x + c_0$. We equate the coefficients of $P(x) = (x - A)Q(x)$ and demand that $|a_i| \in \{1, A\}$:

1. $a_0 = -Ac_0 \implies c_0 = \pm 1$.
2. $a_1 = c_0 - Ac_1$. For $|a_1| \in \{1, A\}$, since $A \geq 3$, it is strictly required that $c_1 = 0$, giving $a_1 = c_0$.
3. $a_2 = c_1 - Ac_2 = -Ac_2$. This forces $c_2 = \pm 1$.
4. $a_3 = c_2 - Ac_3$. Similarly, $c_3 = 0$, giving $a_3 = c_2$.
5. $a_4 = c_3 - Ac_4 = -Ac_4$. This forces $c_4 = \pm 1$.
6. $a_5 = c_4 - Ac_5$. This forces $c_5 = 0$, giving $a_5 = c_4$.
7. $a_6 = c_5 - A = -A$, which correctly gives $|a_6| = A$.

Thus, $Q(x) = x^6 + c_4x^4 + c_2x^2 + c_0$, with $c_4, c_2, c_0 \in \{1, -1\}$. We must also have $P(1) = 0$, meaning $Q(1) = 0$:

$$1 + c_4 + c_2 + c_0 = 0$$

For the sum to be zero, exactly two of c_4, c_2, c_0 must be -1 and one must be 1. The number of valid choices for (c_4, c_2, c_0) is $\binom{3}{1} = 3$.

The value A must be an odd integer strictly greater than 1 and strictly less than 2025^2 . Since $2025^2 = 4100625$, the odd integers allowed are $3, 5, 7, \dots, 4100623$. The number of such odd integers is:

$$\frac{4100625 - 3}{2} = 2050311$$

For each of these A values, there are 3 polynomials, yielding:

$$3 \times 2050311 = 6150933 \text{ polynomials}$$

Total Number of Polynomials Adding the valid polynomials from both cases, we get:

$$6150933 + 35 = 6150968$$

Final Answer There are **6150968** good polynomials.